



Expert Session

An informal expert session on pricing issues

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Agenda

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2. Modifications of insurances
3. Graphical Examples
4. Formal Approach
5. Premium Financing
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Modifications of insurances

New insurances and their modifications

What are we talking about?

- OECD defines how to price new transactions:
 - Minimum pricing methodologies for OECD transactions: MPR, MB
 - ECA-specific pricing methodologies for all other transactions
 - Risk premium is 80% of total premium

However:

- OECD does not help on pricing modifications of insurances
- ECA-specific pricing methodologies for all changes

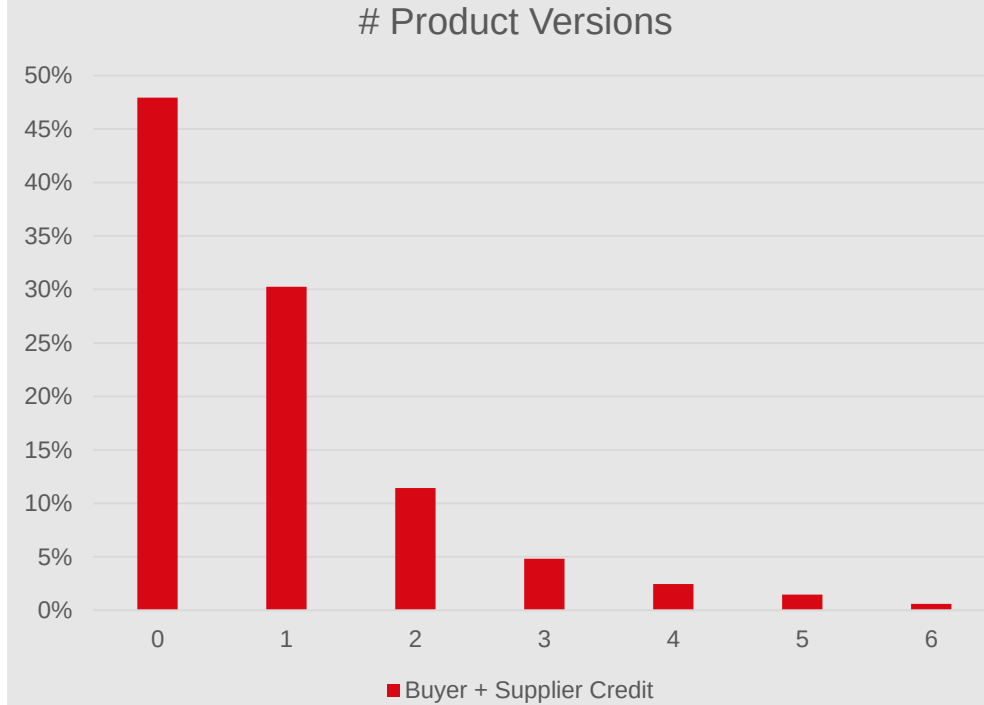


Why is this relevant?

Modifications are frequent: Less than half of the insured buyer + supplier credits stay unmodified.

Why?

- Delivery contract may contain some flexibility in terms of dates (delivery)
- Definition of SPOC might be flexible (commissioning date)
- Delivery is delayed
- Repayments are anticipated
- Repayments are delayed (restructuring)
- Whole transaction is repaid early.



SERV Experience with pricing modifications of insurances

Situation at SERV before 2021:

- No SERV methodology for pricing modifications.
- Modifications were priced as difference to corresponding new transaction.
- With early repayments, administrative premium was reimbursed → net loss

Approach:

- In 2021, SERV started to apply a new calculation method for (full or partial) early repayments or reductions of amounts.
- Since July 2023, SERV applies the same methodology to all modifications.



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Graphical Examples

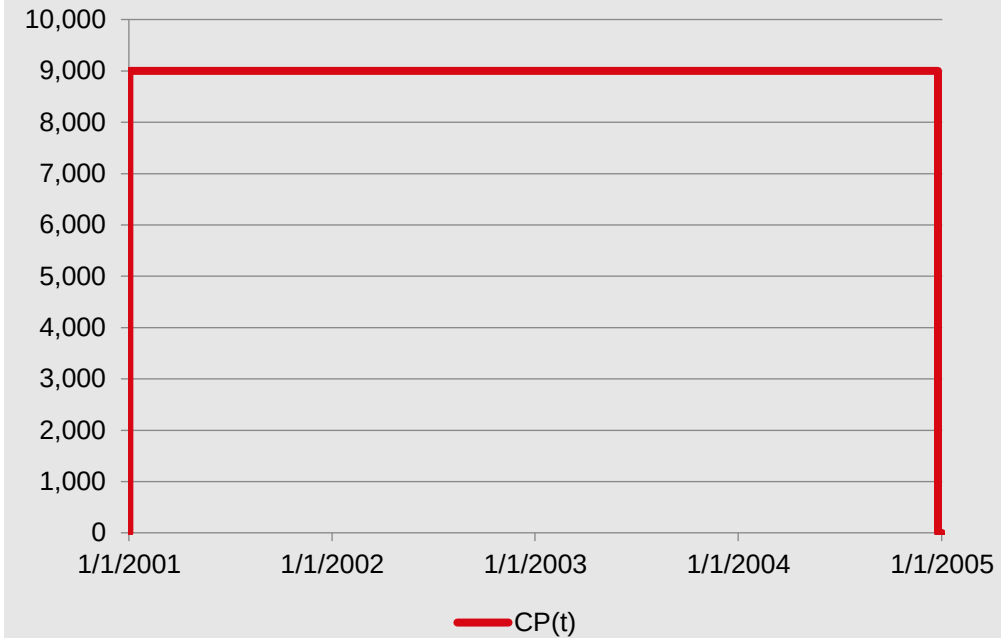
Two graphical examples for calculation

Risk represented by area: Example 1

To show our approach, we use a graphical approach.

As an example, we take the following transaction:

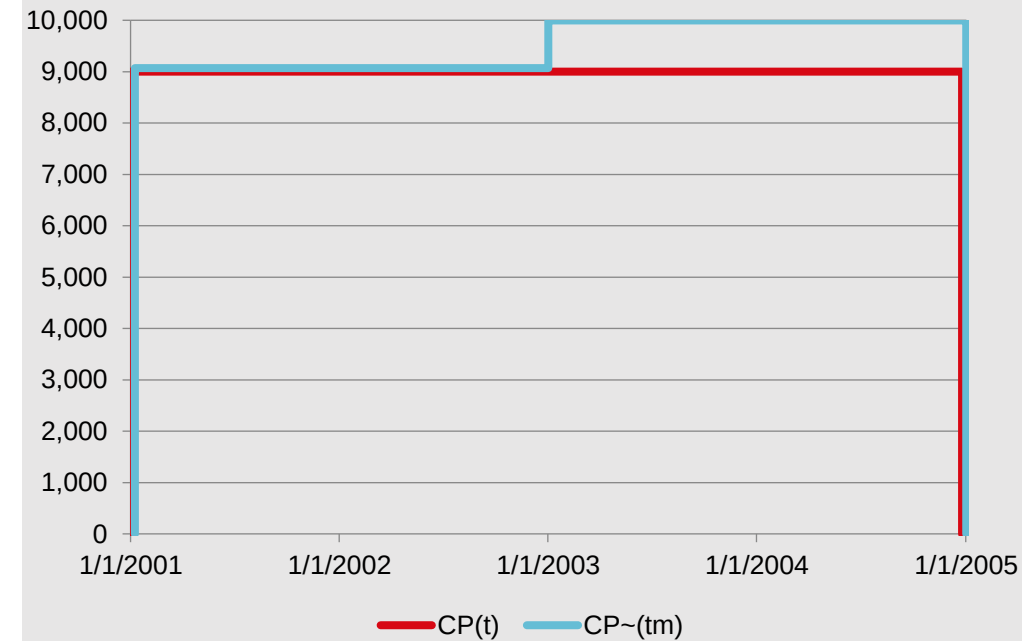
- Buyer Credit
- Cover: 9'000 EUR (might be 9'474 EUR with 95% cover)
- Credit period: 4 years
- Total premium: 2'250 EUR → Risk premium: 1'800 EUR



Risk represented by area: Example 1

As of 1.1.2003, an increase of amount is applied for:

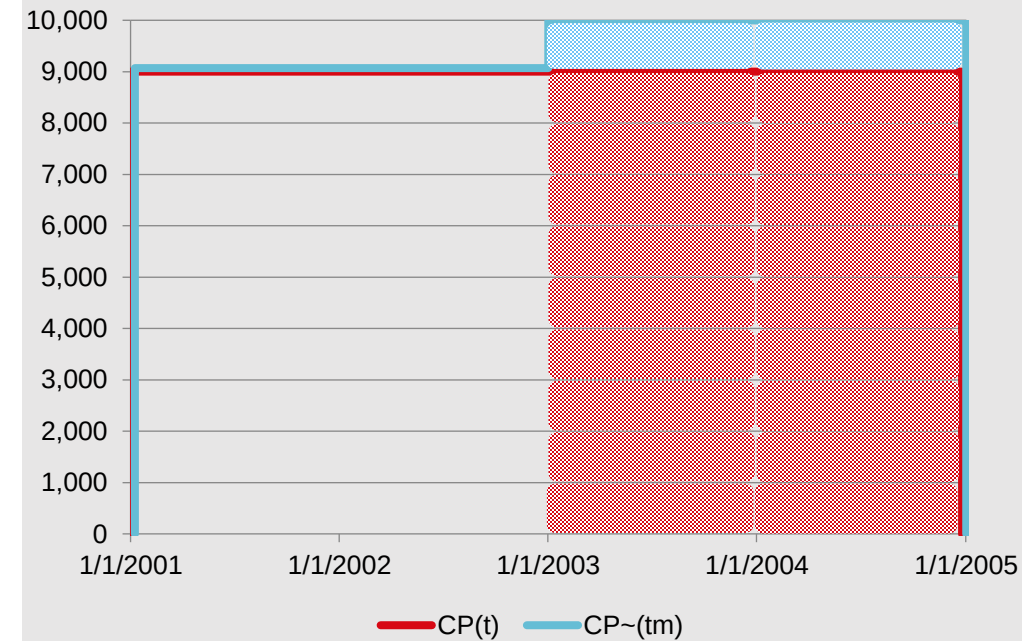
- New Cover: 10'000 EUR
- Date of change: 1.1.2023



Risk represented by area: Example 1

How does the area increase?

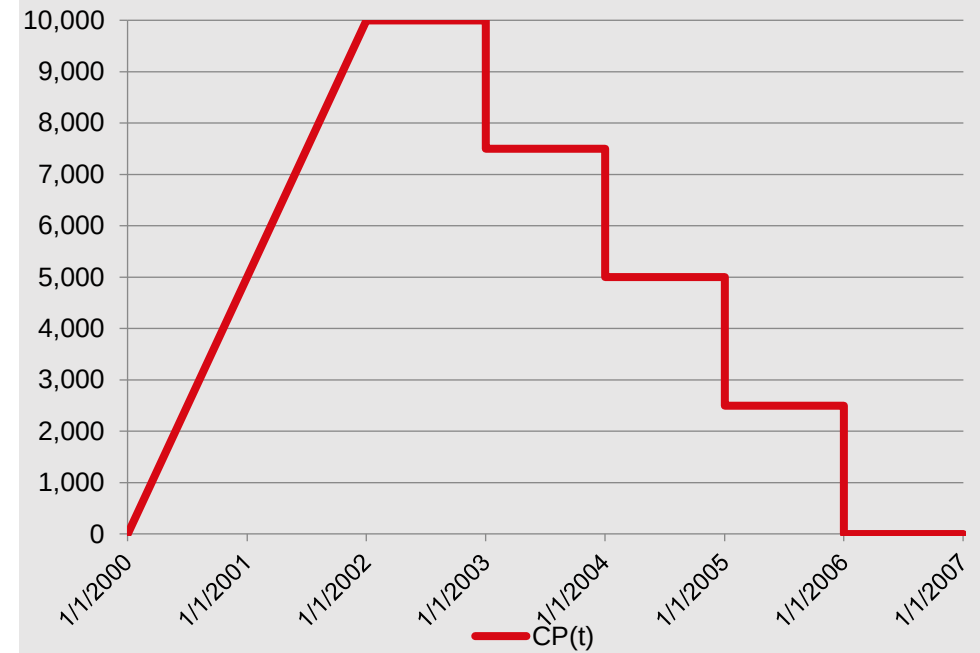
- Red area: 18 «units», corresponds to unearned premium of EUR 900
- Blue area: 2 «units»
- ➔ Increase of Area after 1.1.2003: $2/18=11.1\%$
- ➔ Increase of Risk Premium: $11.1\% \times \text{EUR } 900 = \text{EUR } 100$
- ➔ Increase of Total Premium: $\text{EUR } 100 / 0.8 = \text{EUR } 125$



Risk represented by area: Example 2

Typical structure of Buyer Credit:

- Amortizing Buyer Credit
- Cover: 10'000 EUR (might be 10'526 EUR with 95% cover)
- Disbursement Period: 2 years
- Credit period: 4 years
- Total premium: 2'187 EUR → Risk premium: 1'750 EUR



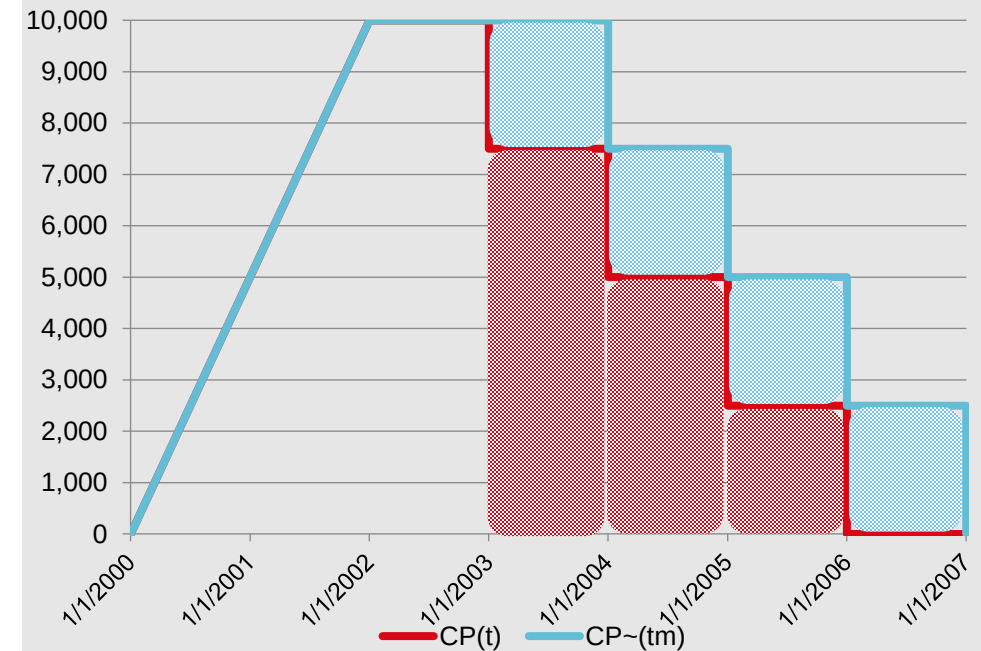
Risk represented by area: Example 2

As of 1.1.2003, a change is applied for:

- Shift of one amortization by one year
- Date of change: 1.1.2003
- Unearned premium: EUR 750

How does the area increase?

- Red area: 6 «units»
- Blue area: 4 «units»
- ➔ Increase of Area after 1.1.2003: $4/6=66.7\%$
- ➔ Increase of Risk Premium: $66.7\% \times \text{EUR } 750 = \text{EUR } 500$
- ➔ Increase of Total Premium: $\text{EUR } 500 / 0.8 = \text{EUR } 625$



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Formal Approach

Establishing a mathematical framework for the pricing of modifications

Specific Risk Premium as Premium Density

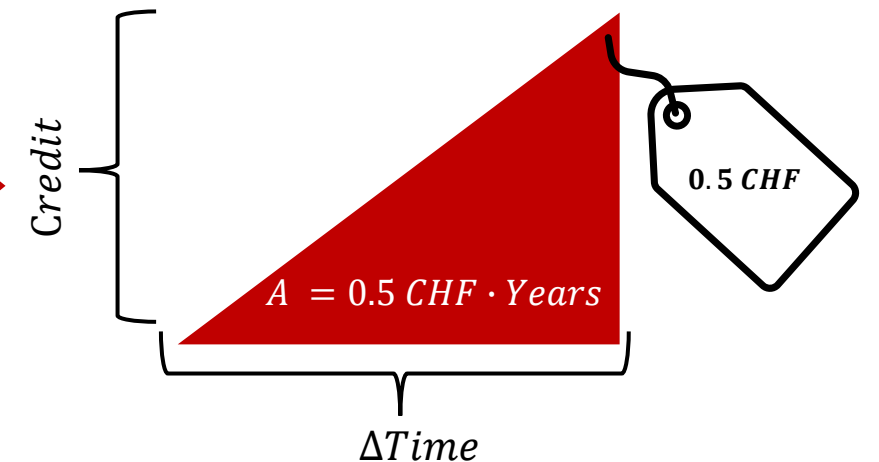
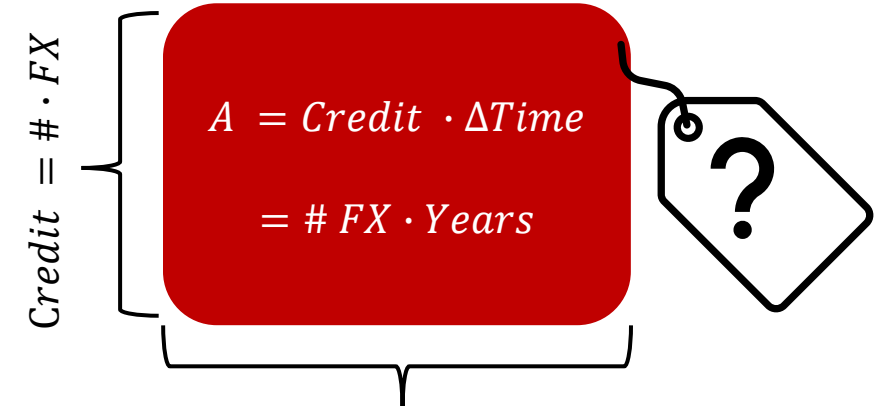
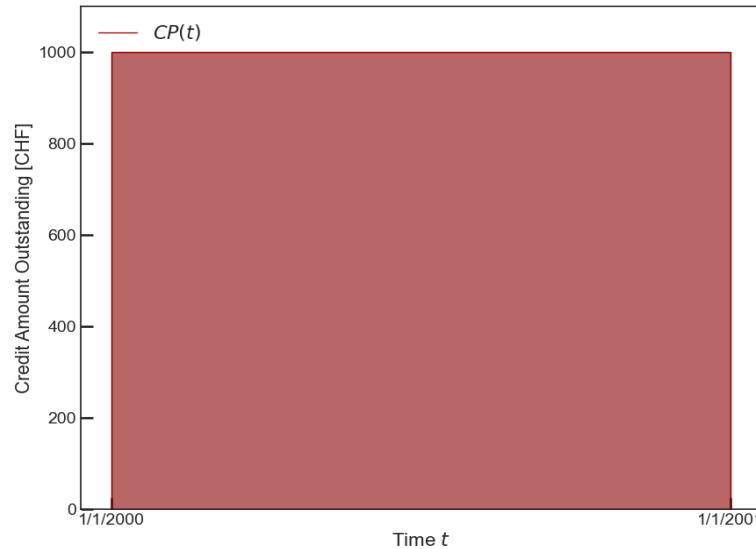
Introduction of the so-called *specific risk premium*:

$$SRP = \frac{RP}{A} = \frac{RP}{\int_{t=-\infty}^{\infty} CP(t) \cdot dt}$$

- RP – risk premium, A – risk area under cover profile $CP(t)$
- SRP quantifies the risk premium due for a risk area of unity

Example

- $RP = 1\,000\text{ CHF}$
- $A = 1\,000\text{ CHF} \cdot \text{Years}$
- $SRP = 1.0 \cdot \frac{1}{\text{Years}}$



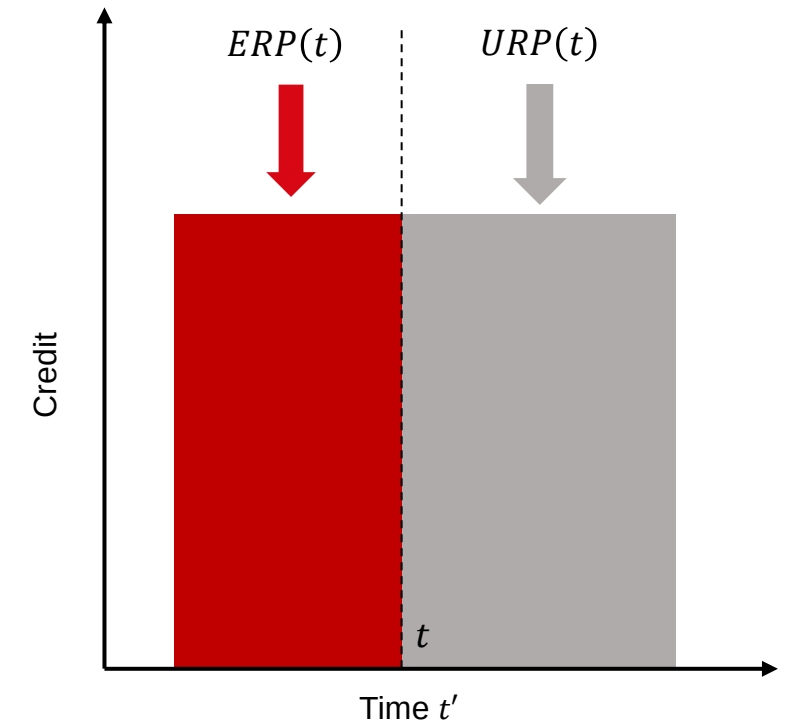
Earned and Unearned Premiums

- The earned risk premium $ERP(t)$ at time t can be defined via the SRP as follows

$$ERP(t) = SRP \cdot \int_{t'=-\infty}^t CP(t') \cdot dt'.$$

- The unearned risk premium $URP(t)$ at time t is the difference of the total risk premium and the amount already earned thereof until time t

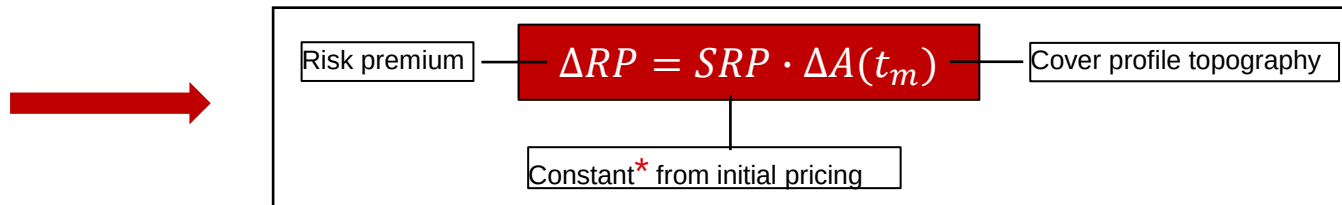
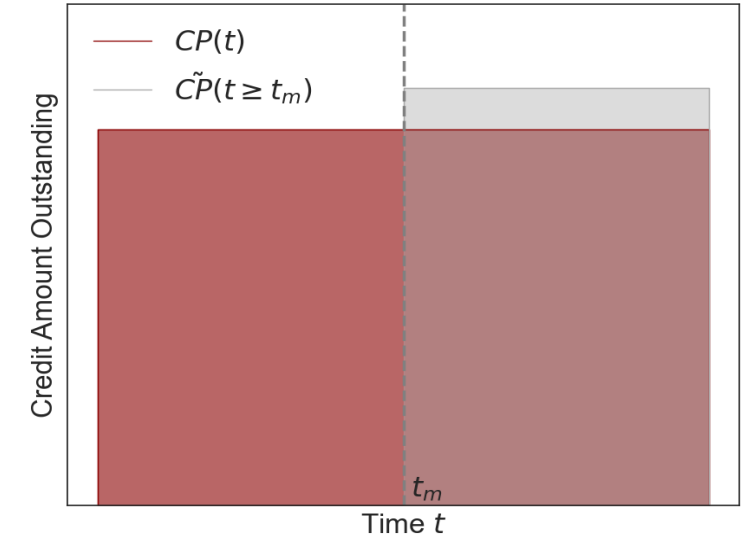
$$URP(t) = RP - ERP(t) = SRP \cdot \int_{t'=t}^{\infty} CP(t') \cdot dt'.$$



The earned and unearned risk premiums are related to the time-sliced cover profile via the specific risk premium and ensure consistency between premium calculation and accrual.

Pricing Framework for Modifications

- $\tilde{CP}(t)$ and $CP(t)$ are the new and old cover profiles corresponding to the initial and modified transaction, respectively
- Let $\tilde{A}(t \geq t_m) = \int_{t=t_m}^{\infty} \tilde{CP}(t) \cdot dt$ and $A(t \geq t_m) = \int_{t=t_m}^{\infty} CP(t) \cdot dt$

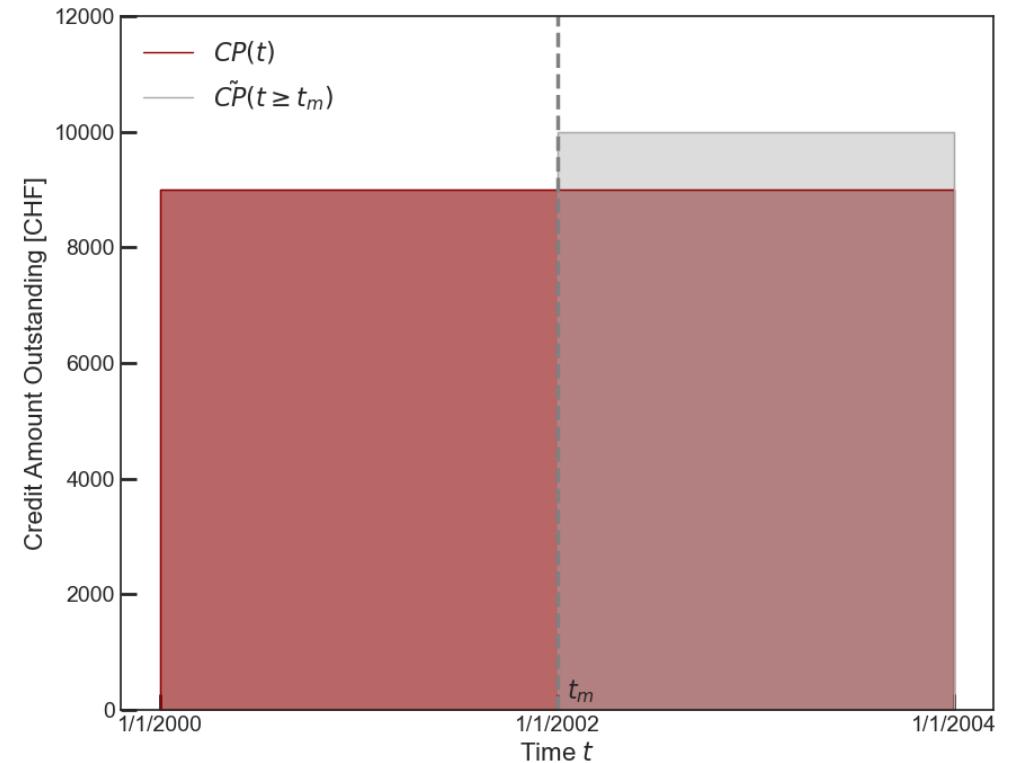


- * Generally, $SRP(t) \neq \tilde{SRP}(t)$.
- In absence of reevaluations (e.g. letter rating updates):
 $SRP(t) = SRP = \tilde{SRP}$

Approach only requires knowledge of the cover profile topography and is therefore universally applicable across different export cases, policy types, etc.

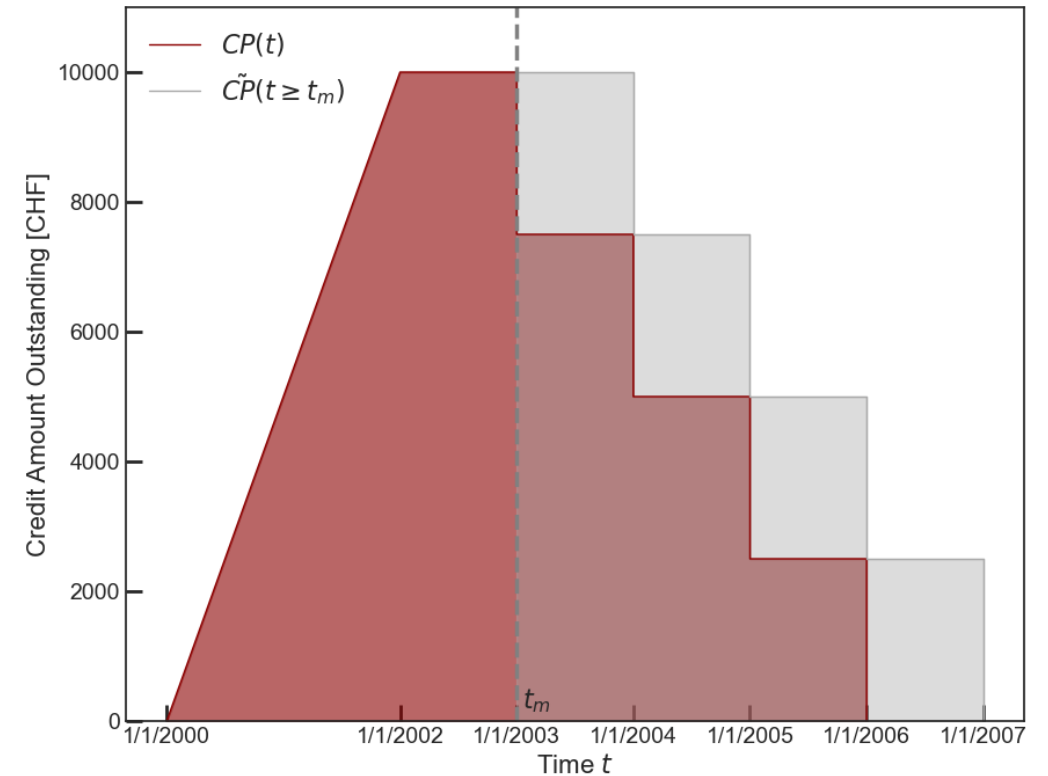
Example I: Increased Risk Coverage

- At $t_m = 1/1/2002$ the covered amount increases from 9000 to 10000 CHF
- The specific risk premium is $SRP = 0.05 \cdot \frac{1}{Years}$



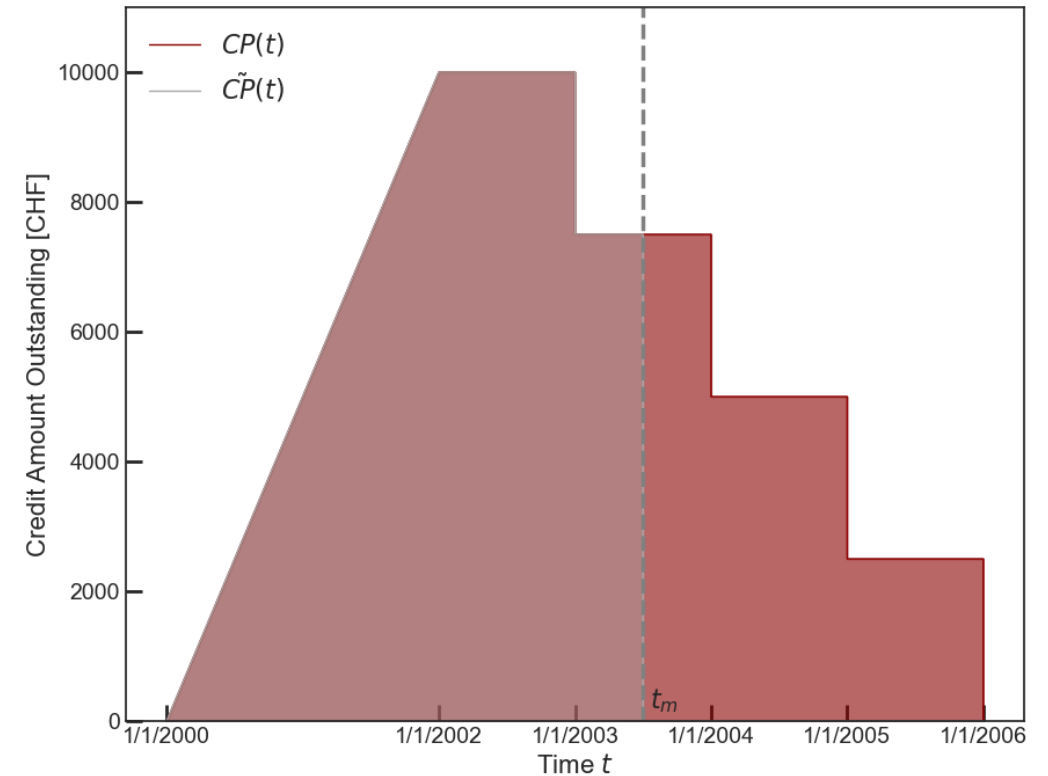
Example II: Extension of Repayment Period

- Deferral of all time-periodic repayments of amount 2500 CHF by one repayment time interval of 1 Year
- The specific risk premium is $SRP = 0.05 \cdot \frac{1}{\text{Years}}$



Example III: Premature Termination

- Termination of an active insurance at time t_m
- The specific risk premium is $SRP = 0.05 \cdot \frac{1}{Years}$



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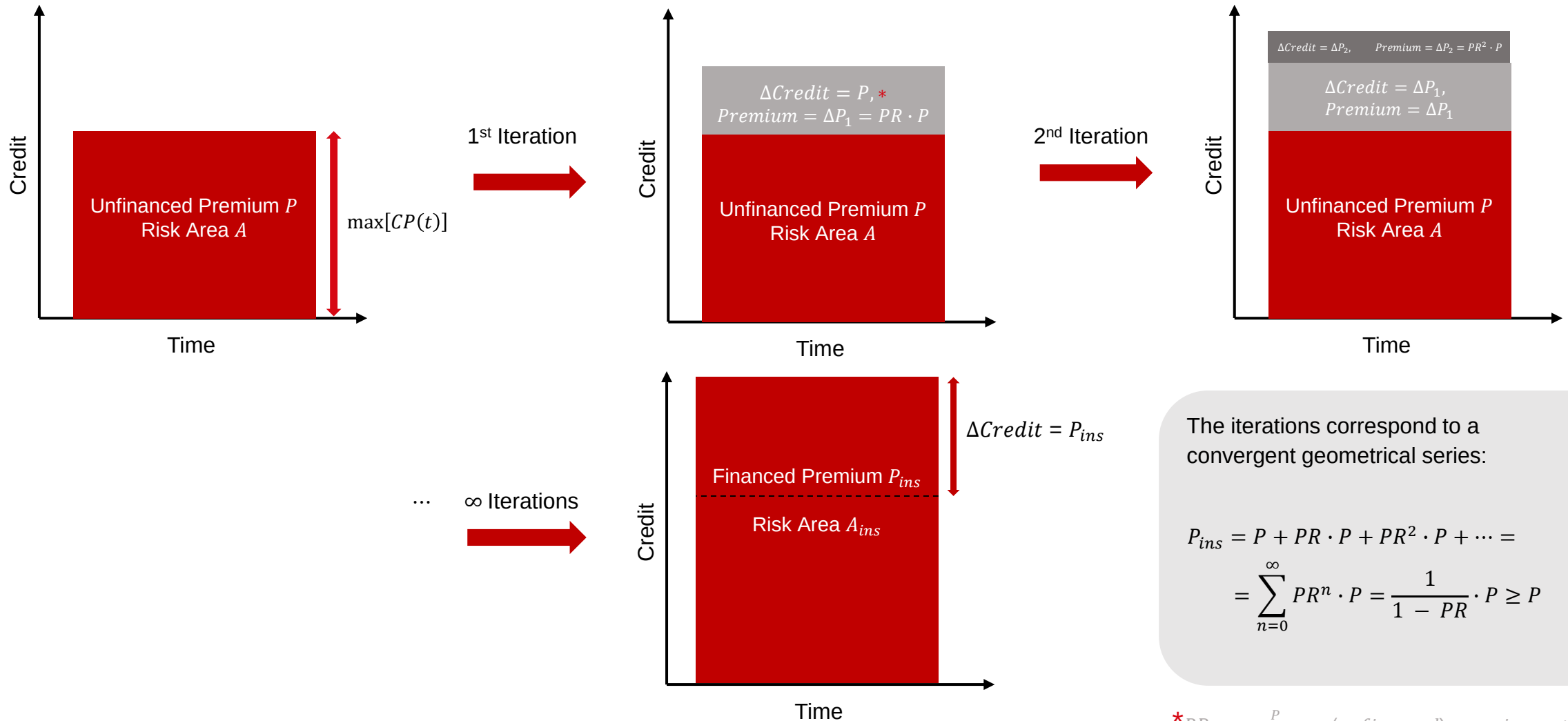
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Premium Financing

Development of a self-consistent solution for the
premium financing of modifications

Recap: Premium Financing



Premium Financing of Modifications

- Financing of a premium surcharge $\Delta P > 0$ corresponds to an increased coverage $\tilde{A}(t \geq t_m)_{ins}$ at the expense of a higher premium $\Delta P_{ins} \geq \Delta P$
- Mathematically, there are two unknown variables:

$$\Delta P_{ins} = \kappa_{\Delta P} \cdot \Delta P \quad (1), \quad \tilde{A}_{ins}(t \geq t_m) = \kappa_{\tilde{A}(t \geq t_m)} \cdot \tilde{A}(t \geq t_m) \quad (2)$$

- Business logic implies two boundary conditions:
 $SRP_{ins} = SRP \quad (1), \quad \max[\tilde{CP}_{ins}(t \geq t_m)] = \max[\tilde{CP}(t \geq t_m)] + \Delta P_{ins} \quad (2)$

OECD results for premium financing are contained as special cases of our solution

In the limit $t_m \rightarrow -\infty$, the solutions for premium financing are recovered:

$$P_{ins} = \frac{1}{1 - PR} \cdot P \geq P,$$

$$A_{ins} = \frac{1}{1 - PR} \cdot A \geq A.$$

Solution:

$$\Delta P_{ins} = \kappa_{\Delta P} \cdot \Delta P = \frac{1}{1 - \frac{SRP \cdot \tilde{A}(t_m)}{\omega \cdot \max[\tilde{CP}(t_m)]}} \cdot \Delta P \geq \Delta P$$

$$\tilde{A}_{ins}(t \geq t_m) = \left[1 + \kappa_{\Delta P} \cdot \frac{\Delta P}{\max[\tilde{CP}(t_m)]} \right] \cdot \tilde{A}(t_m) \geq \tilde{A}(t_m)$$

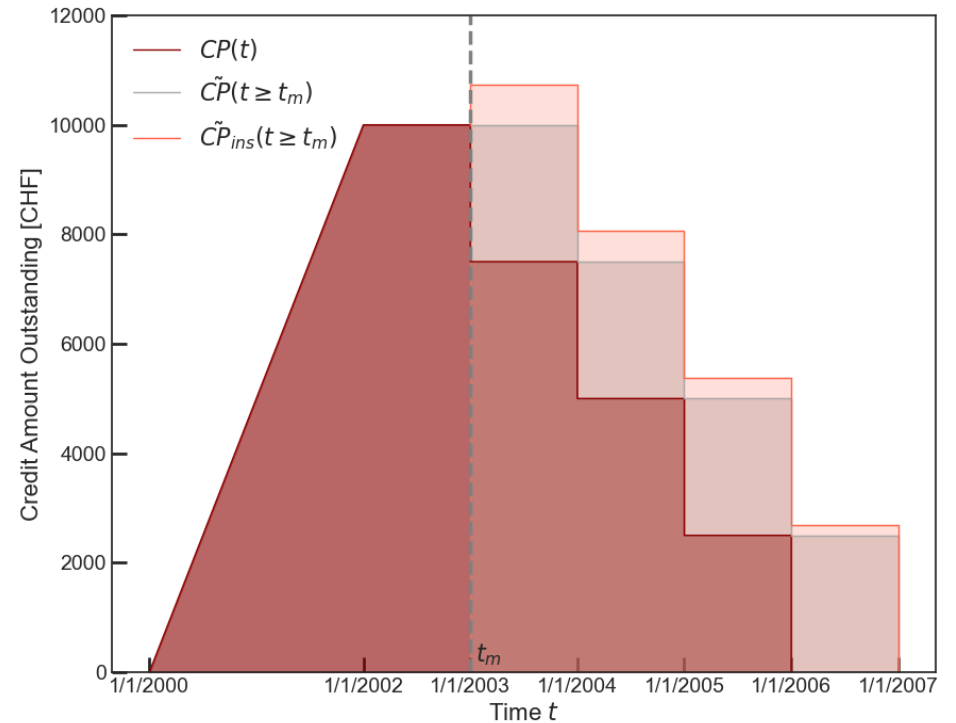
Example II Revisited: Premium Financing

- Same setup as Example II:

$$SRP = 0.05 \cdot \frac{1}{\text{Years}}, \quad \max[\tilde{CP}(t_m)] = 10000 \text{ CHF}$$

- Results without premium financing:

$$\Delta P = 625 \text{ CHF}, \quad \tilde{A}(t_m) = 25000 \text{ CHF} \cdot \text{Years},$$



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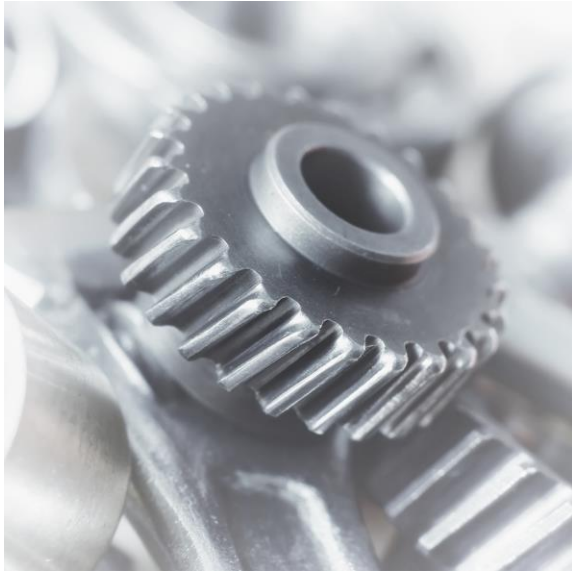
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